Grade 10 – Test 4

Name: _____
Grade / Class: 10

English Language & Literature (15 Questions)

1. **Read the sentence: "Amina completed the mural for the community hall."**
   Question: What did Amina complete?
2. **Identify the error: "He goes to school every days."**
3. **Fill in the blank: "She would have succeeded if she ___ (prepare) thoroughly."**
4. **Rewrite in passive voice: "The team won the championship last year."**
5. **Choose the correct synonym: "Generous" – a) Stingy b) Kind c) Selfish d) Cruel**
6. **Write a short story about a discovery in a forest.**
7. **Read the passage: "The engineer studied different bridges to improve construction."**
   Question: What did the engineer study?
8. **Identify the figure of speech: "The wind whispered secrets through the trees."**
9. **Write a formal letter to request permission to organize a cultural event at school.**
10. **Correct punctuation: "whom did you meet yesterday asked the teacher"**
11. **Fill in the blank: "The project was praised ___ its innovation and creativity."**
12. **Analyze: "Thunder rolled across the valley, echoing like drums."**
    Question: Which literary device is used?
13. **Convert to indirect speech: "I am reading a new book," said Zara.**
14. **Choose the correct word: "He has ___ patience to complete the task." (a) many b) much c) few d) little**
15. **Summarize: "Deforestation leads to loss of biodiversity, soil erosion, and climate change."**

Mathematics (15 Questions)

16. **Solve: 2x2−7x+3=02x^2 - 7x + 3 = 02x2−7x+3=0**
17. **Simplify: (5x2y)(3xy3)(5x^2y)(3xy^3)(5x2y)(3xy3)**
18. **Find the slope of the line passing through points (2,5) and (-3,1).**
19. **Solve the inequality: 4x−9>3x+24x - 9 > 3x + 24x−9>3x+2**
20. **Calculate the area of a trapezium with parallel sides 8 cm and 12 cm and height 5 cm.**
21. **Factorize: x3−27x^3 - 27x3−27**
22. **If f(x)=2x+3f(x) = 2x + 3f(x)=2x+3 and g(x)=x2−1g(x) = x^2 - 1g(x)=x2−1, find f(g(2))f(g(2))f(g(2)).**

23. Find the length of the diagonal of a rectangle with length 6 cm and width 8 cm.
24. Solve for x: 3(x−2)+5=2x+73(x-2) + 5 = 2x + 73(x−2)+5=2x+7
25. Evaluate: 144+33\sqrt{144} + 3^3144+33
26. Draw a graph of y=2x+1y = 2x + 1y=2x+1 for x = -2, -1, 0, 1, 2.
27. Find the probability of drawing a red card from a standard deck of 52 cards.
28. Simplify: 3x2y9xy3\frac{3x^2y}{9xy^3}9xy33x2y
29. A car travels 120 km in 2 hours. Find its speed in km/h.
30. Solve: x2+6x+9=0x^2 + 6x + 9 = 0x2+6x+9=0

Physics (10 Questions)

31. State Newton's Second Law of Motion.
32. A ball is thrown vertically upward with a velocity of 10 m/s. Calculate the maximum height.
33. Define work done in physics.
34. A force of 5 N moves a block 2 m. Calculate the work done.
35. Distinguish between scalar and vector quantities.
36. Explain Ohm's Law.
37. A circuit has a 12 V battery and a resistance of 4 Ω. Find the current.
38. Define kinetic energy and potential energy.
39. Give one real-life example of simple harmonic motion.
40. What is the difference between speed and velocity?

Chemistry (10 Questions)

41. Define atomic number and mass number.
42. Explain the difference between ionic and covalent bonds.
43. Balance the chemical equation: H2+O2→H2O\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}H2+O2→H2O
44. Name two acids and two bases commonly used in daily life.
45. Explain the pH scale.
46. Define oxidation and reduction.
47. What are isotopes? Give one example.
48. Differentiate between endothermic and exothermic reactions.
49. Calculate the molar mass of CO2\text{CO}_2CO2.
50. State one use of catalysts in chemical reactions.

Biology (10 Questions)

51. Explain the structure of a human heart.
52. Define photosynthesis and write its chemical equation.
53. Differentiate between aerobic and anaerobic respiration.
54. What is osmosis? Give one example.
55. Name the main parts of a neuron.
56. Define chromosomes.
57. Explain the role of white blood cells in the human body.
58. Describe the human digestive system briefly.
59. Differentiate between plant and animal cells.
60. What are vaccines and why are they important?

Social Studies / History / Geography (10 Questions)

61. Define democracy and name two countries practicing it.
62. Explain the causes of World War I.
63. Name five continents and their largest countries.
64. What is GDP and why is it important for a country?
65. Define renewable and non-renewable resources.
66. Explain the greenhouse effect and its consequences.
67. Who was Nelson Mandela and why is he famous?
68. Name two major rivers in Asia and their importance.
69. Define the term “globalization.”
70. Explain the main features of the Cold War.

Islamiyat / Moral Education (10 Questions)

71. Name the five pillars of Islam.
72. What is Zakat and its importance in society?
73. Define Salah and mention its five daily timings.
74. Explain the concept of fasting (Sawm) during Ramadan.
75. Write briefly about the life of Prophet Muhammad (PBUH).
76. Define Halal and Haram with one example each.
77. What is the importance of honesty in Islam?
78. Explain the significance of Hajj.
79. Mention two important lessons from the Quran.
80. Define Jihad in its true Islamic sense.

Computer Science / ICT (10 Questions)

81. Define algorithm and give a simple example.
82. Differentiate between hardware and software.
83. What is a database and give one use.
84. Name three programming languages commonly used today.
85. Explain the difference between RAM and ROM.
86. Define binary system.
87. What is the function of an operating system?
88. Explain the difference between the internet and intranet.
89. Write the main steps of problem-solving in computing.
90. Define artificial intelligence and one real-life application.

Economics / Business Studies (10 Questions)

91. Define supply and demand with one example each.
92. Explain the concept of inflation.
93. Name two types of business organizations.
94. What is a market economy?
95. Define entrepreneurship.
96. What is the role of banks in an economy?
97. Differentiate between profit and revenue.
98. Explain the term “opportunity cost.”
99. Name two global stock exchanges.
100. Define international trade and give one advantage.